

Canada based PhD candidate in Computational and Cognitive Neuroscience at the Hebrew University of Jerusalem, focusing on visual attention in naturalistic conditions. Former software engineer at Facebook (Meta), with a BSc. in Computer Science at Tel Aviv University. I am proficient in eye-tracking and EEG analysis, statistics, data science and software engineering. I have a proven ability to lead complex projects from planning through execution, both as a team leader and a collaborative team member.

Education

- 2022-present **Hebrew University (PhD Candidate): [ELSC Direct PhD Track](#) in Computational Neuroscience**
Advisor: Prof. Leon Deouell, MD PhD
- Course GPA: 95.8
- 2022-2025 **Hebrew University (MSc): Computational Neuroscience (Direct PhD Track)**
Advisor: Prof. Leon Deouell, MD PhD
- Final GPA: 90.1
 - Developed and published an open-source Python package for benchmarking eye-tracking data parsers.
 - Experienced in co-registration and analysis of large-scale multi-modal physiological and behavioral data.
- 2016-2020 **Tel-Aviv University (BSc): Computer-Science & Psychology (Emphasis on Neuroscience)**
- Final GPA: 86.5
 - BSc. Thesis: *A no-report paradigm for probing human memory using eye tracking*.
Advisors: Omer Sharon, PhD; Prof. Yuval Nir, PhD
 - BSc. Thesis: *Face recognition during free viewing of naturalistic video stimuli*.
Advisor: Prof. Galit Yovel, PhD
 - Studied a semester in Konstanz University, Germany (Mar-Aug 2018).
- 2007-2010 **"Ironi Hey" High School (Haifa)**
- Majored in Math, English, Physics and Chemistry at the highest level.

Publications

- 2026
- Nir, J. (2026). Data Sharing and Citation Impact in Visual Search Research. *PsyArXiv*, 2026-04. [pre-print]
 - Nir, J. & Deouell, L.Y. (2026). Systematic Classification Differences Across Eye-Movement Detection Algorithms. *Behavior Research Methods* 58, 109.
- 2025
- Schmidig, F. J., ..., Nir, J., ... & Nir, Y. (2025). Anticipatory eye gaze as a marker of memory. *Communications Psychology*, 3(1), 122.

Conferences

- 2026
- Nir, J., Auerbach-Asch R.C., & Deouell, L.Y. (2026, May 15-19). *Fixated but Failed to See: Recognition Errors in Visual Search*. Annual Meeting of the Vision Science Society ([VSS 2026](#)), St. Pete Beach, FL, USA. [poster presentation]
- 2025
- Nir, J. & Deouell, L.Y. (2025, Jul. 6-9). *pEYES: An Open-Source Python Toolkit for Eye Tracking Data Analysis and Detector Evaluation*. Association for the Scientific Study of Consciousness Annual Meeting ([ASSC 28](#)), Heraklion, Greece. [poster presentation]
 - Nir, J. & Deouell, L.Y. (2025, Feb. 25-27). *A Comparative Evaluation of Eye Movement Detection Algorithms*. Israeli Society for Cognitive Psychology Annual Conference ([ISCOP 11](#)), Acre, Israel. [poster presentation]

Awards & Scholarships

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| 2022-present | • ELSC Excellence Scholarship for Direct-Track PhD Students (info). |
| 2024 | • Israeli National Road Safety Authority (רלב"ד) research grant for MSc students (info). |
| | • Israeli Smart Transportation Research Center (ISTRC) scholarship for MSc students (info). |
| 2018 | • Erasmus+ scholarship for Israeli students studying in European universities (info). |
| 2015 | • Military-Intelligence's Major-General award for Creative Thinking. |

Professional Experience

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| 2022-2023 | Academic Teaching: Hebrew University <ul style="list-style-type: none">• Teaching Assistant: Summer course for graduate students in neuroscience from the Hebrew University & Technical University of Munich (76953; September 2023, HUJI+TUM)• Lab Instructor: Introduction to Machine Learning (67577; spring semester 2023, HUJI)• Teaching Assistant: Statistical Methods (51307; winter semester 2022, HUJI) |
| 2019-2022 | Software Engineer: Express Wi-Fi (Facebook TLV) <ul style="list-style-type: none">• Unblocked deployment in multiple regions by integrating our product (multi-service, AWS) with FB's internal SMS platform, while maintaining data separation required for GDPR compliance.• Increased ad-impressions by 30% by adjusting existing customer-flows and supporting new ad-types.• Team PoC for experimentation processes: design, implementation & review. |
| 2017-2019 | Research Assistant & Software Developer: Yuval Nir Lab (Tel-Aviv University) <ul style="list-style-type: none">• Implemented industry-standard feature-extraction algorithms for eye-tracking data (python).• Conducted clinical trials on humans to collect physiological measures, such as EEG, eye tracking, etc. |

Military Service

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| 2015-2016 | Signal Intelligence Research Team Leader in Unit 8200 (Lieutenant) <ul style="list-style-type: none">• Military-Intelligence's Major-General award for Creative Thinking (2015).• Lead a multi-disciplinary intelligence research group, as part of an international cooperation framework.• Accompanied technological development teams: from product characterization (PRDs), through monitoring implementation progress to overseeing deployment.• Participated in long term planning: allocating manpower and resources, project prioritization, and predicting future requirements.• Direct people management of a team of 5-8 researchers.• Worked in a highly variant and demanding conditions, requiring fast adjustments & multi-tasking. |
| 2013-2015 | Signal Intelligence Analyst Officer (קב"ר) in Unit 8200 |
| 2011-2013 | Intelligence Analyst in Unit 8200 |

Languages and Skills

Example Projects

- **Hebrew** – native; **English** – fluent.
- Programming languages: python, Go, javascript & React.
- Experiment planning, execution and data analysis.
- Fast learning, independent and responsible.
- Good presentational & interpersonal skills.
- Hobbies: Yoga, CrossFit, Hiking, Cooking.
- *pEYES*, a package for eye-tracking algorithm evaluation and analysis [python; [github](#)]
- Implementing SLAM on KITTI dataset using visual odometry, bundle adjustment & gtsam [python; [github](#)]